Hypothalamus, Thalamus, and Pituitary Gland

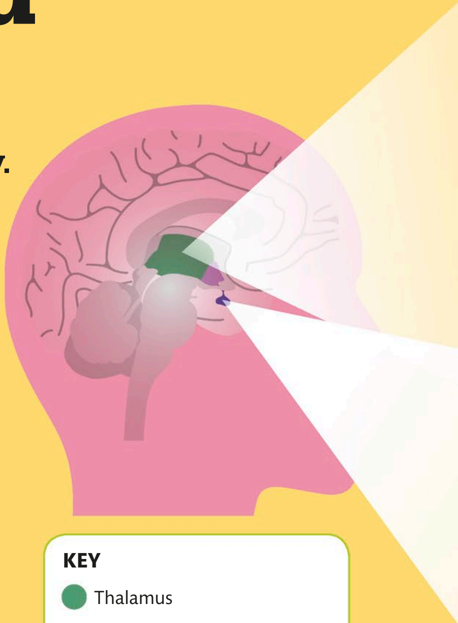
The thalamus and the structures around it sit at the center of the brain. They act as relay stations between the forebrain and the brain stem, also forming a link to the rest of the body.

The hypothalamus

This small region under the forward region of the thalamus is the main interface between the brain and the hormone, or endocrine, system. It does this by releasing hormones directly into the bloodstream, or by sending commands to the pituitary gland to release them. The hypothalamus has a role in growth, homeostasis (maintaining optimal body conditions), and significant behaviors such as eating and sex. This makes it responsive to many different stimuli.

WHAT GLANDS DOES THE PITUITARY GLAND CONTROL?

The pituitary gland is a master gland that controls the thyroid gland, adrenal gland, ovaries, and testes. However, it receives its instructions from the hypothalamus.

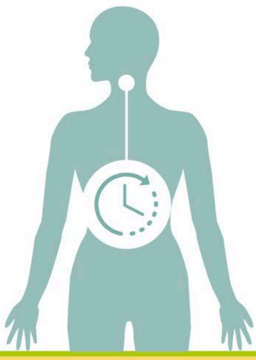


KEY

- Thalamus
- Hypothalamus
- Pituitary gland

THE EPITHALAMUS

This small region covers the top of the thalamus. It contains various nerve tracts that form a connection between the forebrain and midbrain. It is also the location of the pineal gland—the source of melatonin, a hormone central to the sleep-wake cycle and body clock.



RESPONSES OF THE HYPOTHALAMUS

STIMULUS	RESPONSE
Day length	Helps maintain body rhythms after receiving signals about day length from the optical system.
Water	When the blood's water levels drop, releases vasopressin, also called antidiuretic hormone, which reduces the volume of urine.
Eating	When the stomach is full, releases leptin to reduce feelings of hunger.
Lack of food	When the stomach is empty, releases ghrelin to boost feelings of hunger.
Infection	Increases body temperature to help the immune system work faster to fight off pathogens.
Stress	Increases the production of cortisol, a hormone associated with preparing the body for a period of physical activity.
Body activity	Stimulates the production of thyroid hormones to boost the metabolism, and somatostatin to reduce it.
Sexual activity	Organizes the release of oxytocin, which helps the formation of interpersonal bonds. The same hormone is released during childbirth.